

# DevelopersHub Corporation

## AI/ML Engineering – Advanced Internship Tasks

**Due Date:** 28th April, 2026

### Overview

As part of your **AI/ML Engineering Internship** at **DevelopersHub Corporation**, you are required to complete **at least 3 out of the 5 advanced tasks** listed below within the given timeline.

These tasks are carefully designed to give you hands-on experience with **cutting-edge machine learning and artificial intelligence techniques**, including **transformer models, ML pipelines, multimodal learning, conversational AI, and LLM applications**.

You are encouraged to complete all tasks to strengthen your technical portfolio. Technologies and tools you'll use include:

Hugging Face Transformers, scikit-learn, LangChain, Streamlit, Gradio, CNNs, joblib, and LLMs.

### Advanced Task Set

#### Task 1: News Topic Classifier Using BERT

**Objective:**

Fine-tune a transformer model (e.g., BERT) to classify news headlines into topic categories.

**Dataset:**

AG News Dataset (Available on Hugging Face Datasets)

**Instructions:**

- Tokenize and preprocess the dataset
- Fine-tune the `bert-base-uncased` model using Hugging Face Transformers
- Evaluate the model using accuracy and F1-score

- Deploy the model using **Streamlit** or **Gradio** for live interaction

#### **Skills Gained:**

- NLP using Transformers
- Transfer learning & fine-tuning
- Evaluation metrics for text classification
- Lightweight model deployment

## **Task 2: End-to-End ML Pipeline with Scikit-learn Pipeline API**

#### **Objective:**

Build a reusable and production-ready machine learning pipeline for predicting customer churn.

#### **Dataset:**

Telco Churn Dataset

#### **Instructions:**

- Implement data preprocessing steps (e.g., scaling, encoding) using **Pipeline**
- Train models like Logistic Regression and Random Forest
- Use **GridSearchCV** for hyperparameter tuning
- Export the complete pipeline using **joblib**

#### **Skills Gained:**

- ML pipeline construction
- Hyperparameter tuning with GridSearch

- Model export and reusability
- Production-readiness practices

### **Task 3: Multimodal ML – Housing Price Prediction Using Images + Tabular Data**

**Objective:**

Predict housing prices using both structured data and house images.

**Dataset:**

Housing Sales Dataset + Custom Image Dataset (your own or any public source)

**Instructions:**

- Use CNNs to extract features from images
- Combine extracted image features with tabular data
- Train a model using both modalities
- Evaluate performance using MAE and RMSE

**Skills Gained:**

- Multimodal machine learning
- Convolutional Neural Networks (CNNs)
- Feature fusion (image + tabular)
- Regression modeling and evaluation

### **Task 4: Context-Aware Chatbot Using LangChain or RAG**

**Objective:**

Build a conversational chatbot that can remember context and retrieve external information during conversations.

**Dataset:**

Custom corpus (e.g., Wikipedia pages, internal documents, or any knowledge base)

**Instructions:**

- Use LangChain or Retrieval-Augmented Generation (RAG)
- Implement context memory for conversational history
- Retrieve answers from a vectorized document store
- Deploy the chatbot with **Streamlit**

**Skills Gained:**

- Conversational AI development
- Document embedding and vector search
- Retrieval-Augmented Generation (RAG)
- LLM integration and deployment

**Task 5: Auto Tagging Support Tickets Using LLM****Objective:**

Automatically tag support tickets into categories using a large language model (LLM).

**Dataset:**

Free-text Support Ticket Dataset **Instructions:**

- Use prompt engineering or fine-tuning with an LLM • Compare zero-shot vs fine-tuned performance
- Apply few-shot learning techniques to improve accuracy
- Output top 3 most probable tags per ticket

### **Skills Gained:**

- Prompt engineering
- LLM-based text classification
- Zero-shot and few-shot learning
- Multi-class prediction and ranking

## **Submission Requirements (Per Task)**

Each completed task must be **uploaded to your GitHub repository** and submitted via **Google Classroom**.

### **Checklist:**

#### **1. Jupyter Notebook / Script File**

Problem Statement & Objective

- Dataset Loading & Preprocessing
- Model Development & Training
- Evaluation with relevant metrics
- Visualizations (if applicable)
- Final Summary / Insights

#### **2. Code Quality**

0 Clear structure, logical flow, comments explaining major steps

#### **3. GitHub Repository**

Clear repository name and organization ○ **README.md** must include:

- Objective of the task

- Methodology / Approach
- Key results or observations

#### 4. Submission on Google Classroom

Submit the GitHub repo link for each completed task

## Important Notes

- **Deadline:** Complete at least **3 out of 5 tasks** by **28th April, 2026**
- You are encouraged to complete all 5 for advanced portfolio building
- Share your work on GitHub and LinkedIn to showcase your skills